

NAME

sigprocmask, rt_sigprocmask – examine and change blocked signals

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <signal.h>

/* Prototype for the glibc wrapper function */
int sigprocmask(int how, const sigset_t *_Nullable restrict set,
                sigset_t *_Nullable restrict oldset);

#include <signal.h>      /* Definition of SIG_* constants */
#include <sys/syscall.h>  /* Definition of SYS_* constants */
#include <unistd.h>

/* Prototype for the underlying system call */
int syscall(SYS_rt_sigprocmask, int how,
            const kernel_sigset_t *_Nullable set,
            kernel_sigset_t *_Nullable oldset,
            size_t sigsetsize);

/* Prototype for the legacy system call */
[[deprecated]] int syscall(SYS_sigprocmask, int how,
                           const old_kernel_sigset_t *_Nullable set,
                           old_kernel_sigset_t *_Nullable oldset);
```

Feature Test Macro Requirements for glibc (see **feature_test_macros(7)**):

```
sigprocmask():
    _POSIX_C_SOURCE
```

DESCRIPTION

sigprocmask() is used to fetch and/or change the signal mask of the calling thread. The signal mask is the set of signals whose delivery is currently blocked for the caller (see also **signal(7)** for more details).

The behavior of the call is dependent on the value of *how*, as follows.

SIG_BLOCK

The set of blocked signals is the union of the current set and the *set* argument.

SIG_UNBLOCK

The signals in *set* are removed from the current set of blocked signals. It is permissible to attempt to unblock a signal which is not blocked.

SIG_SETMASK

The set of blocked signals is set to the argument *set*.

If *oldset* is non-NULL, the previous value of the signal mask is stored in *oldset*.

If *set* is NULL, then the signal mask is unchanged (i.e., *how* is ignored), but the current value of the signal mask is nevertheless returned in *oldset* (if it is not NULL).

A set of functions for modifying and inspecting variables of type *sigset_t* ("signal sets") is described in **sigsetops(3)**.

The use of **sigprocmask()** is unspecified in a multithreaded process; see **pthread_sigmask(3)**.

RETURN VALUE

sigprocmask() returns 0 on success. On failure, *-1* is returned and *errno* is set to indicate the error.

ERRORS**EFAULT**

The *set* or *oldset* argument points outside the process's allocated address space.

EINVAL

Either the value specified in *how* was invalid or the kernel does not support the size passed in *sigsetsize*.

VERSIONS**C library/kernel differences**

The kernel's definition of *sigset_t* differs in size from that used by the C library. In this manual page, the former is referred to as *kernel_sigset_t* (it is nevertheless named *sigset_t* in the kernel sources).

The glibc wrapper function for **sigprocmask()** silently ignores attempts to block the two real-time signals that are used internally by the NPTL threading implementation. See **nptl(7)** for details.

The original Linux system call was named **sigprocmask()**. However, with the addition of real-time signals in Linux 2.2, the fixed-size, 32-bit *sigset_t* (referred to as *old_kernel_sigset_t* in this manual page) type supported by that system call was no longer fit for purpose. Consequently, a new system call, **rt_sigprocmask()**, was added to support an enlarged *sigset_t* type (referred to as *kernel_sigset_t* in this manual page). The new system call takes a fourth argument, *size_t sigsetsize*, which specifies the size in bytes of the signal sets in *set* and *oldset*. This argument is currently required to have a fixed architecture specific value (equal to *sizeof(kernel_sigset_t)*).

The glibc **sigprocmask()** wrapper function hides these details from us, transparently calling **rt_sigprocmask()** when the kernel provides it.

STANDARDS

POSIX.1-2008.

HISTORY

POSIX.1-2001.

NOTES

It is not possible to block **SIGKILL** or **SIGSTOP**. Attempts to do so are silently ignored.

Each of the threads in a process has its own signal mask.

A child created via **fork(2)** inherits a copy of its parent's signal mask; the signal mask is preserved across **execve(2)**.

If **SIGBUS**, **SIGFPE**, **SIGILL**, or **SIGSEGV** are generated while they are blocked, the result is undefined, unless the signal was generated by **kill(2)**, **sigqueue(3)**, or **raise(3)**.

See **sigsetops(3)** for details on manipulating signal sets.

Note that it is permissible (although not very useful) to specify both *set* and *oldset* as NULL.

SEE ALSO

kill(2), **pause(2)**, **sigaction(2)**, **signal(2)**, **sigpending(2)**, **sigsuspend(2)**, **pthread_sigmask(3)**, **sigqueue(3)**, **sigsetops(3)**, **signal(7)**